

IT3091 – Machine Learning

Lecture 8

Unsupervised learning I: clustering and validation

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Learning Aims

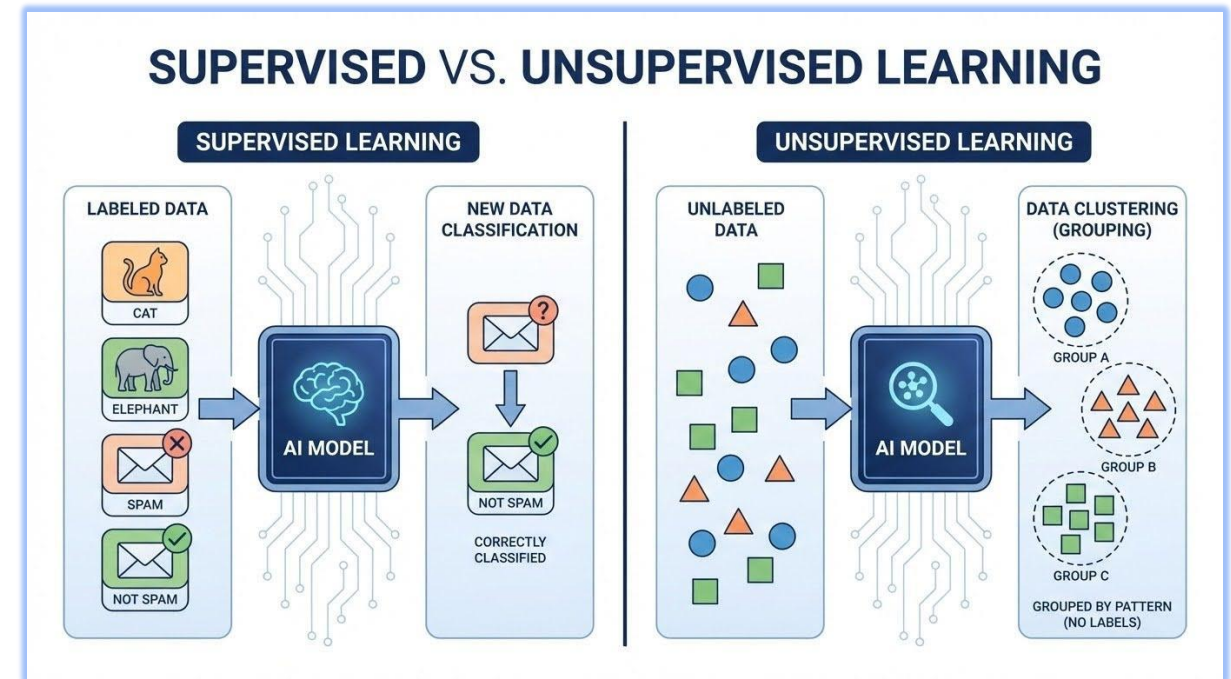
By the end of this lecture, students will be able to:

- ▶ Explain the purpose and key concepts of unsupervised learning and clustering.
- ▶ Apply similarity and distance measures and explain the importance of feature scaling in clustering.
- ▶ Describe and apply K-Means clustering, including centroid assignment, initialization, and choosing K.
- ▶ Explain hierarchical clustering, dendrograms, linkage methods, and cluster cutting.
- ▶ Describe DBSCAN and explain density-based clustering, including the identification of dense regions and noise/outliers.
- ▶ Compare K-Means, hierarchical clustering, and DBSCAN based on their clustering approaches and resulting cluster structures.
- ▶ Evaluate clustering solutions and determine whether they represent meaningful structure rather than noise.
- ▶ Interpret cluster characteristics and relate them to a business or research problem.
- ▶ Compare alternative clustering solutions based on stability, interpretability, and usefulness.
- ▶ Describe a practical clustering workflow from data preparation and distance selection through validation and interpretation.

Introduction to Unsupervised Learning

- ▶ Unsupervised learning finds patterns or structures in data without a known response/output variable.

Supervised Learning	Unsupervised Learning
Target/output available	No target/output
Learn from labelled data	Learn from unlabeled data
Predict an outcome	Discover structure/patterns



Importance

Unsupervised learning allows us to explore datasets when the desired output or labels are not available.

Discovering Hidden Patterns

Main concept

Instead of asking
“What is the output?”

Unsupervised learning asks
“What structure or pattern exists in the data?”

What can we discover

Groups

- Similar observations may form natural groups.

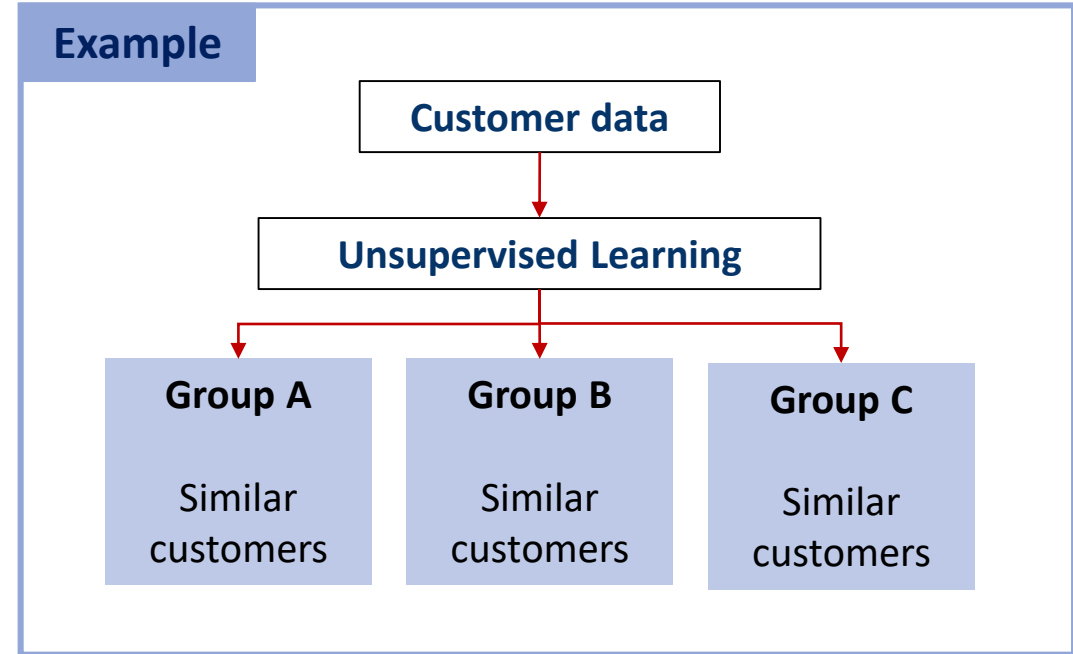
Relationships

- Features may show meaningful associations.

Hidden Structure

- Patterns may exist that are not obvious from raw data.

Example



Key Takeaway

No Target → Discover Patterns

Clustering Fundamentals

- ▶ Clustering is the process of grouping observations that are similar to each other.

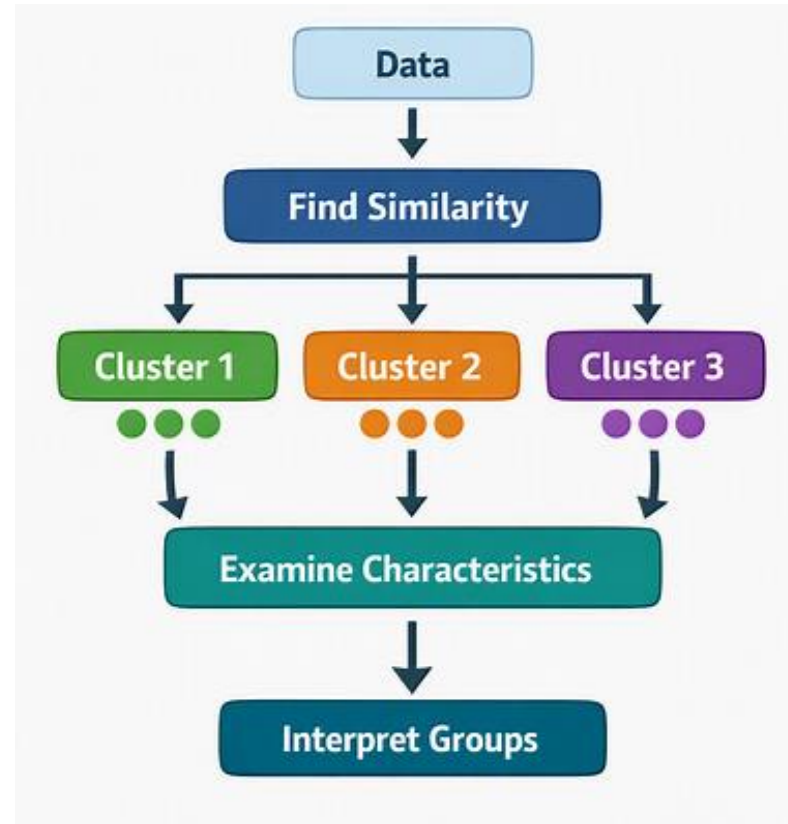
Core Idea

Similar observations → Same cluster
Different observations → Different clusters

What does clustering identify

Groups of similar observations
Natural subgroups
Structure within the data

Finding clusters is only the first step;
understanding what each cluster represents
makes the analysis useful.



Cluster Interpretation

After identifying clusters, check:
“What characteristics make the
observations in this cluster similar?”

Key takeaway

Similarity → Groups → Subgroups → Interpretation

Similarity & Distance Measures

Euclidean Distance

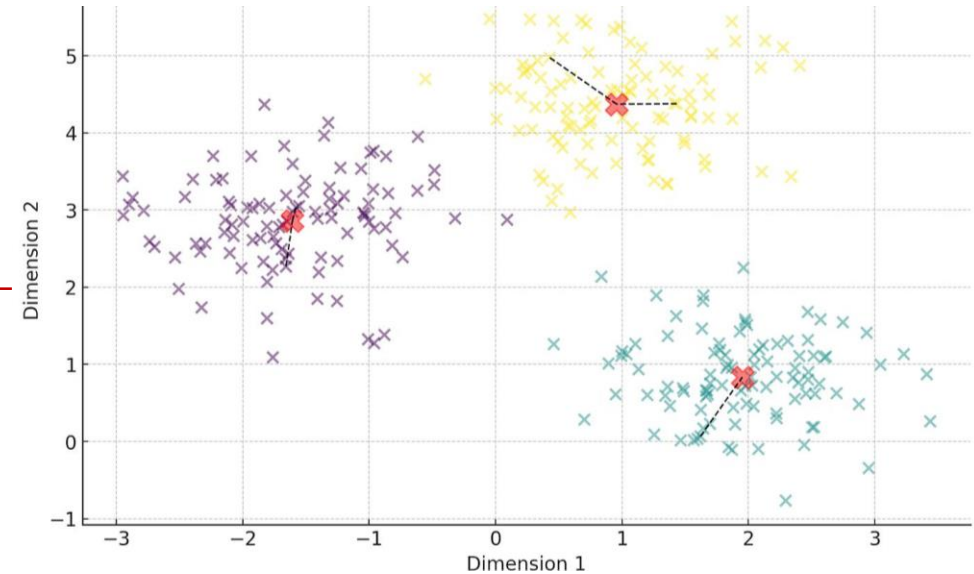
For two observations

$$A = (x_1, x_2), \quad B = (y_1, y_2)$$

$$d(A, B) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

For multiple features

$$d(A, B) = \sqrt{\sum_{j=1}^p (x_j - y_j)^2}$$



Importance

Clustering algorithms use similarity or distance to determine which observations should belong to the same group.

Interpretation

Small distance → Observations are more similar
Large distance → Observations are more different

Choosing a Distance Measure

Data Type	Possible Approach
Numerical	Euclidean distance
Binary / Categorical	Appropriate categorical similarity/distance
Mixed data	Distance measure designed for mixed variables

Important considerations

Before choosing a distance measure consider

- Type of data
- Scale of features
- Meaning of the variables
- Problem context

Feature Scale Matters

Age → 20 – 60

Income → 20,000 – 200,000

Income has a much larger numerical scale and can dominate Euclidean distance.

Key takeaway

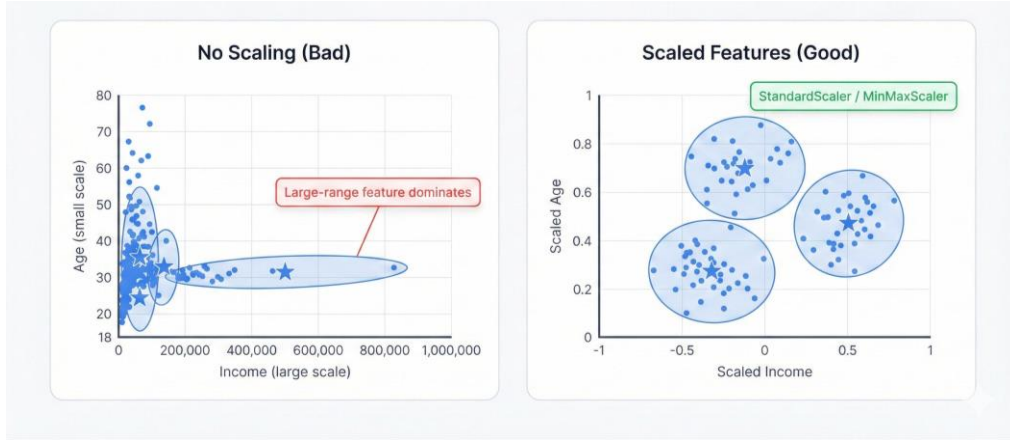
Data Type + Scale + Problem → Choose Distance Measure

Importance

The choice of distance measure can strongly affect which observations are considered similar and, therefore, the clusters that are formed.

Feature Scaling for Clustering

- Features with larger numerical scales can dominate distance calculations and influence the resulting clusters.



Effect of Scaling on Clustering

Without Scaling	After Scaling
Large-scale features can dominate	Features have comparable influence
Distance becomes biased toward those features	Distance reflects multiple features more fairly
Clustering results may be misleading	Clusters can better represent overall similarity

Common approach - Standardization

$$z = \frac{x - \mu}{\sigma}$$

x = original value

μ = feature mean

σ = feature standard deviation

Key takeaway

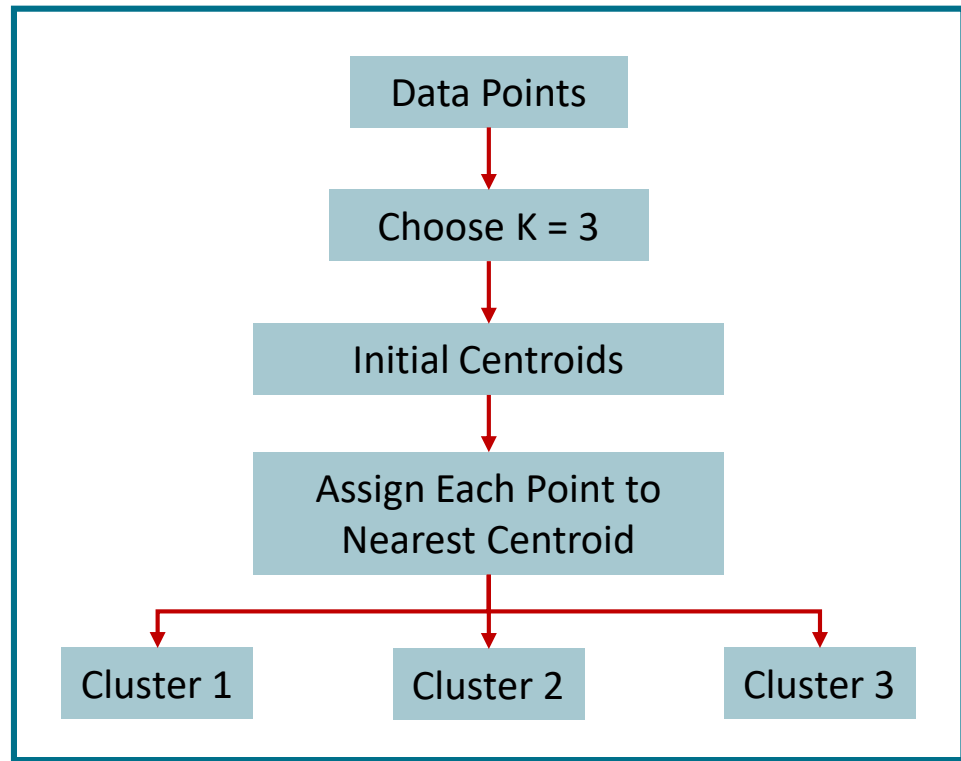
For distance-based clustering, scale features appropriately before clustering when feature ranges differ substantially.

Workflow

Raw Data → Scale Features → Calculate Distances → Cluster

K-Means Clustering

- K-Means is an unsupervised learning method that divides observations into K clusters based on similarity



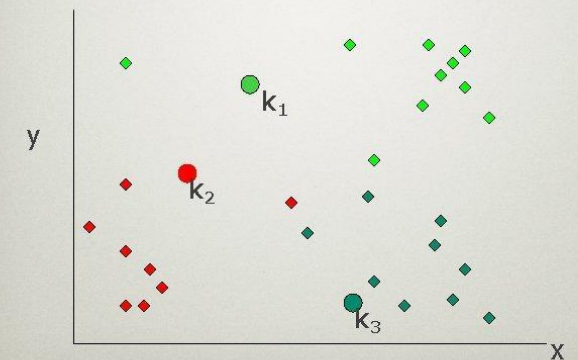
Key Concepts

- **K** → Number of clusters to create
- **Centroid** → Center/mean position of a cluster
- **Cluster Assignment** → Each observation is assigned to the nearest centroid

Why K-Means

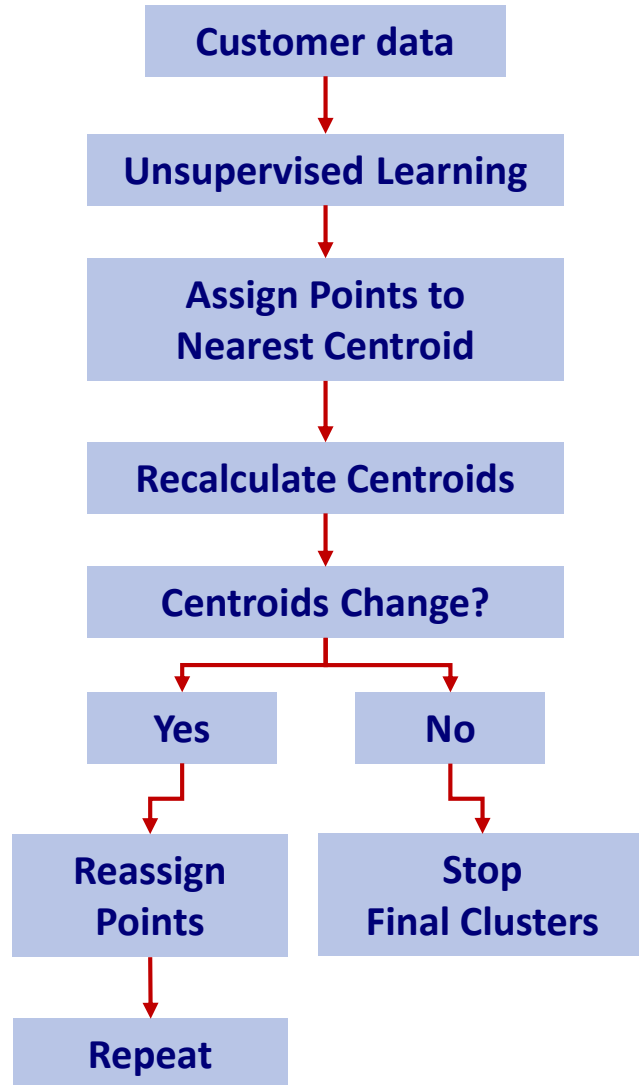
To discover natural groups in data when the group labels are unknown.

K-means clustering (k=3)



Assign each point to nearest cluster center

How K-Means Works



Within-Cluster Variation

K-Means tries to make observations within each cluster as similar as possible.

$$\text{Within-Cluster Variation} = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

C_k = cluster k
 x_i = observation
 μ_k = centroid of cluster k

Goal

Minimize Within-Cluster Variation

Key takeaway

K-Means repeatedly assigns observations to the nearest centroid and updates the centroids until the clusters stabilize.

K-Means Practical Considerations

Random Initialization

Initial centroids are chosen randomly.
Different starting centroids can produce different clusters

Local Optimum

K-Means may stop at a **local optimum** rather than the best possible clustering.
The final result depends on the initial centroid positions.

Multiple Initial Configurations (*n_init*)

Run K-Means multiple times with different initializations.
Select the solution with the lowest within-cluster variation

Reproducibility

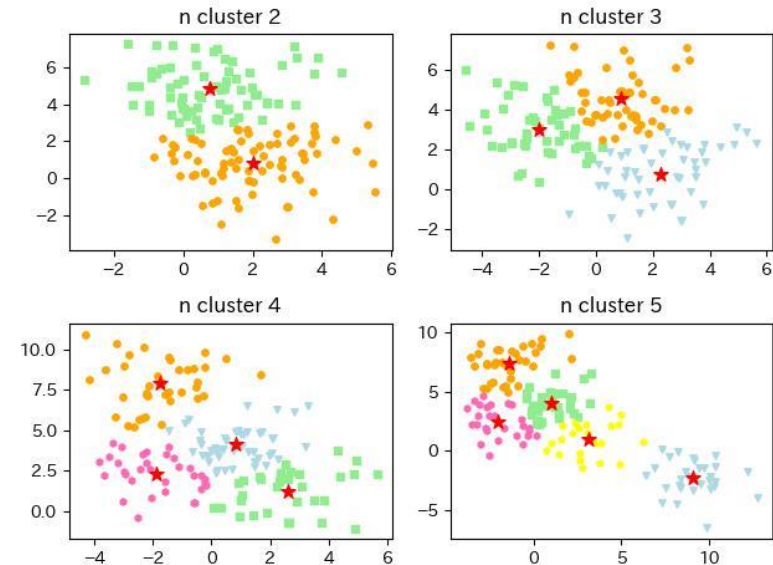
Use a fixed random seed to obtain consistent results.

Choosing the Value of K

Show the same dataset clustered with:

$K = 2 \rightarrow K = 3 \rightarrow K = 4 \rightarrow K = 5$

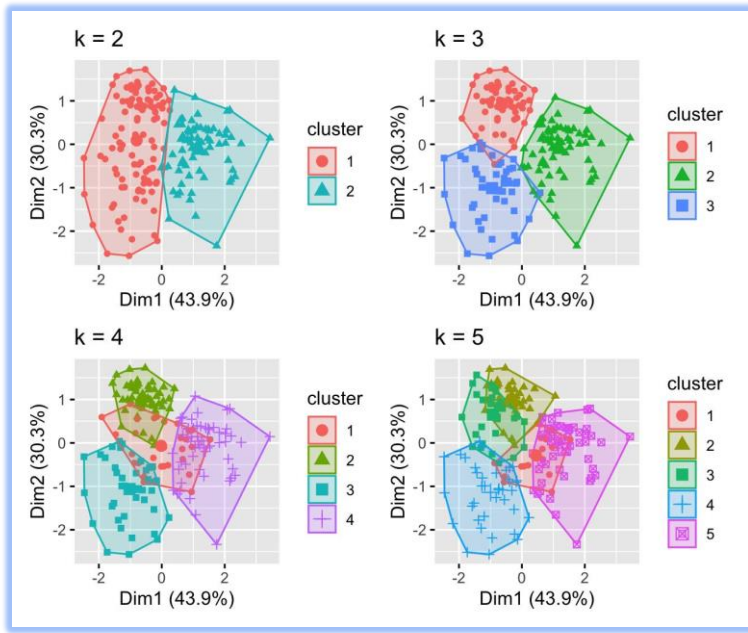
- Small K: may combine distinct groups.
- Large K: may create unnecessary/small clusters.
- The choice of K affects the final clustering structure.



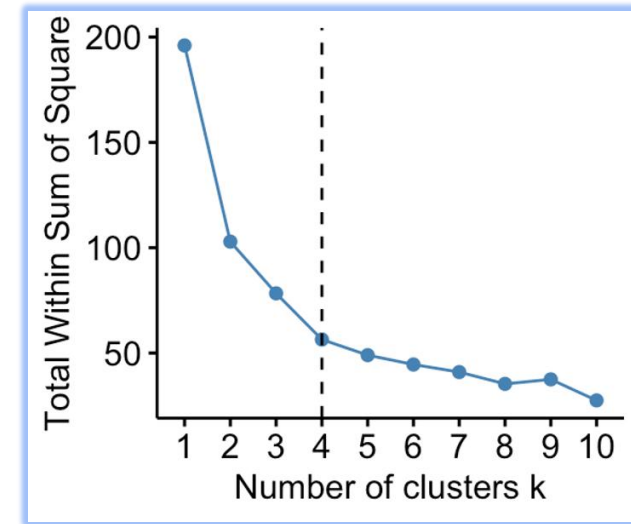
Selecting the Number of Clusters

- ▶ K-Means requires us to choose the number of clusters K before clustering.

- Run K-Means with different values of K
- Compare the resulting clustering solutions
- Select a K that provides a meaningful grouping of the data



Elbow Approach

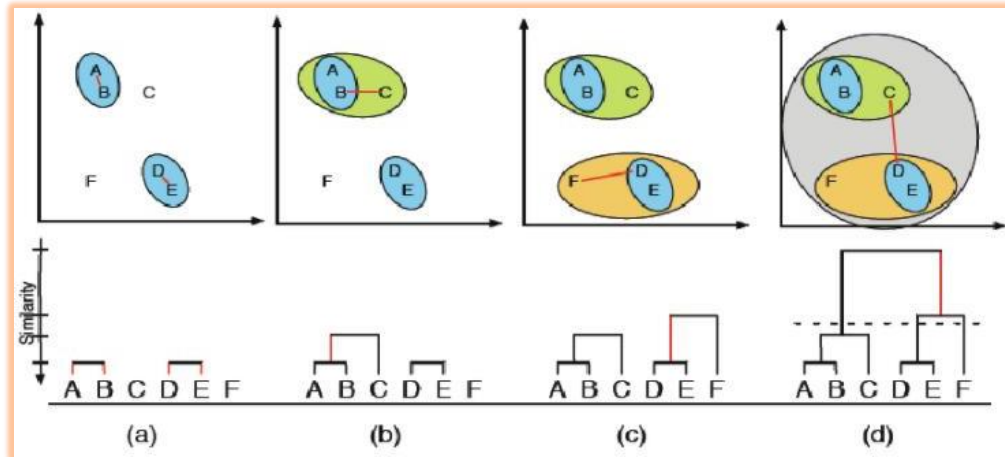


- Compare clustering performance for different K values.
- Within-cluster variation decreases as K increases.
- Improvement becomes smaller after a certain point.
- The elbow provides a heuristic for selecting a reasonable K .

Hierarchical / Agglomerative Clustering

- Hierarchical clustering builds a hierarchy of clusters by progressively merging similar observations

Agglomerative = Bottom-Up



- Start with each observation as its own cluster
- Find the closest clusters
- Merge them
- Repeat until all observations form one cluster

Simple visual flow

Individual Points → Small Clusters → Larger Clusters → One Cluster

Why use

Reveals the hierarchical structure of the data
Does not require choosing K before building the hierarchy

Dendrogram & Fusion Heights

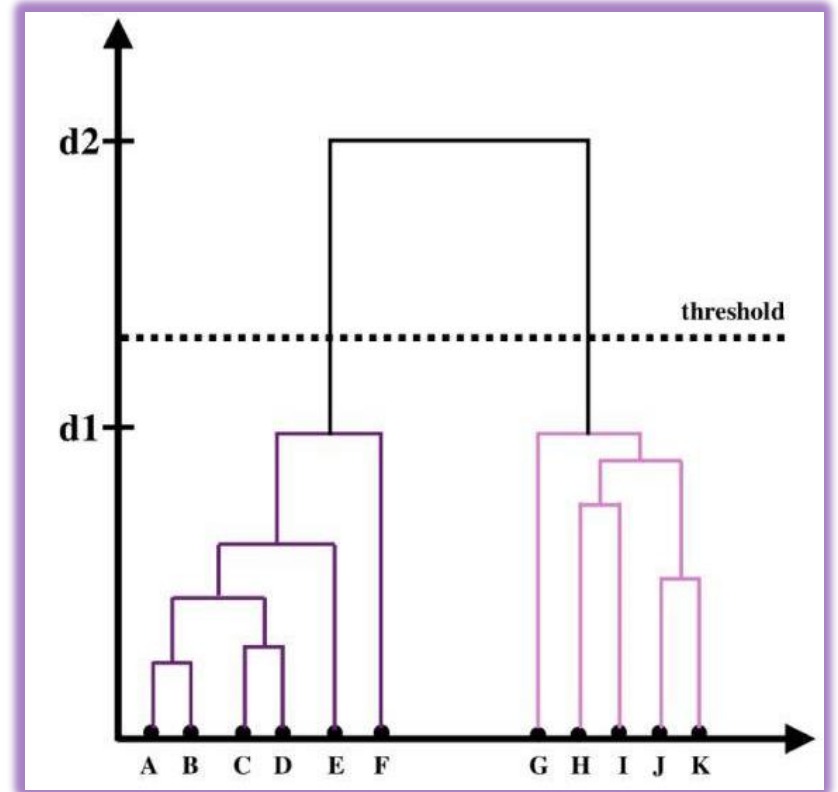
- ▶ A dendrogram is a tree-like diagram that shows how observations/clusters are progressively merged.

How to interpret it

- Leaves → individual observations.
- Branches → cluster mergers.
- Fusion height → distance/dissimilarity at which two clusters are merged.
- Higher fusion height → clusters being merged are more dissimilar.

Key takeaway

Cut the dendrogram at a chosen height to obtain clusters.



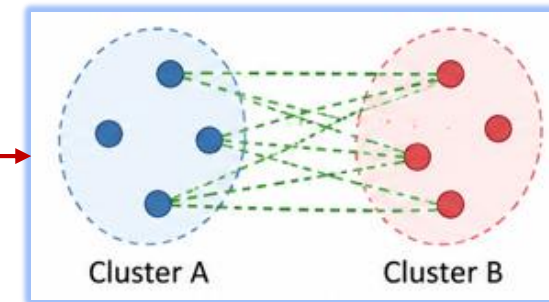
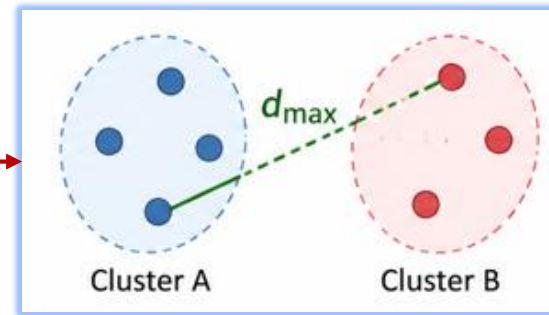
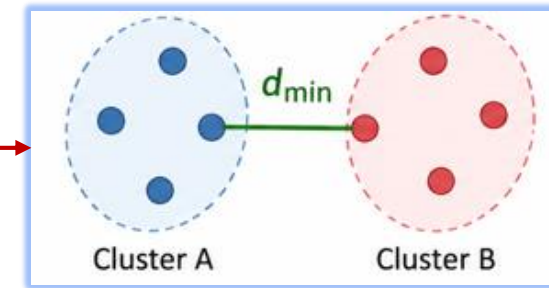
Hierarchical Clustering Linkage

- ▶ Linkage determines the distance between two clusters when deciding which clusters to merge.

Linkage	Cluster distance
Single linkage	Distance between the closest pair of observations
Complete linkage	Distance between the farthest pair of observations
Average linkage	Average distance between observations in the two clusters

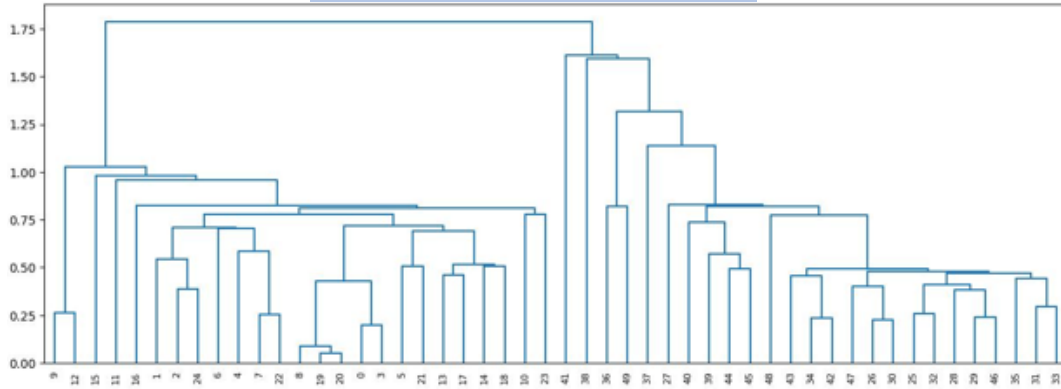
Key idea

Different linkage methods can produce different clustering structures

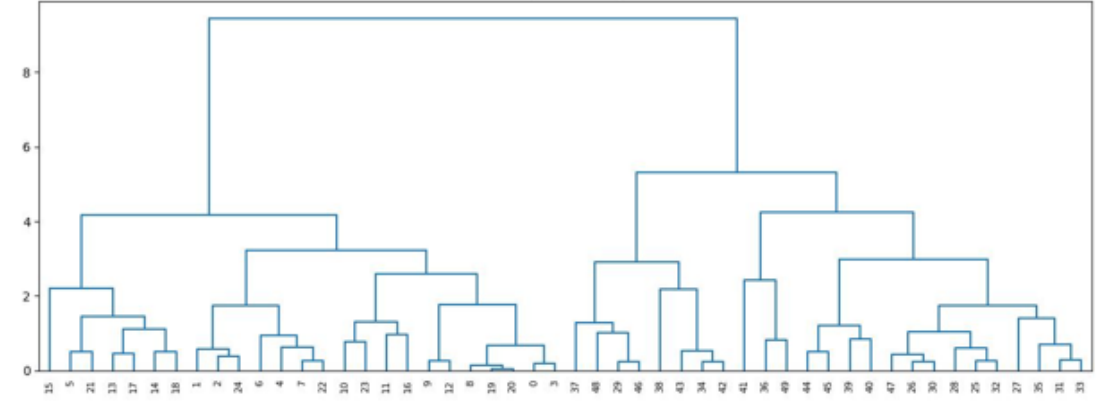


Effect of Linkage Choice

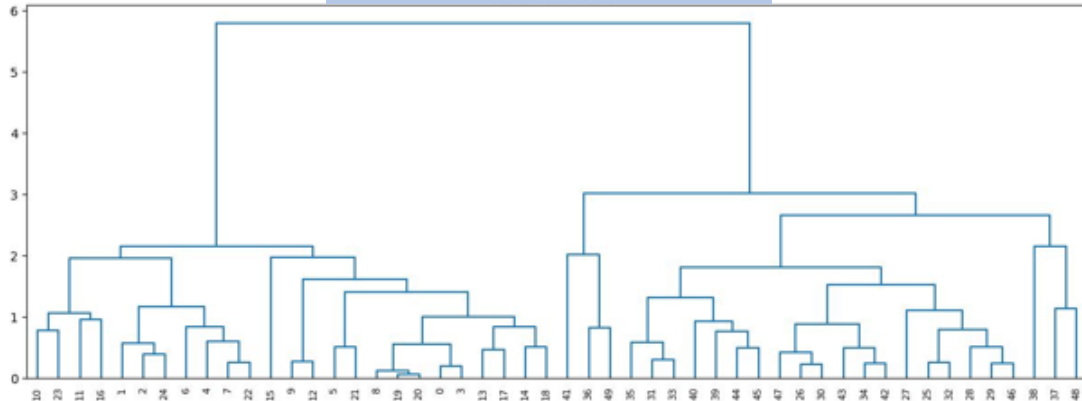
Single Linkage



Complete Linkage



Average Linkage



What changes

- The order of cluster merging can change
- The dendrogram structure can change
- The resulting clusters can therefore be different

Interpreting and Cutting Dendrograms

Dendrogram

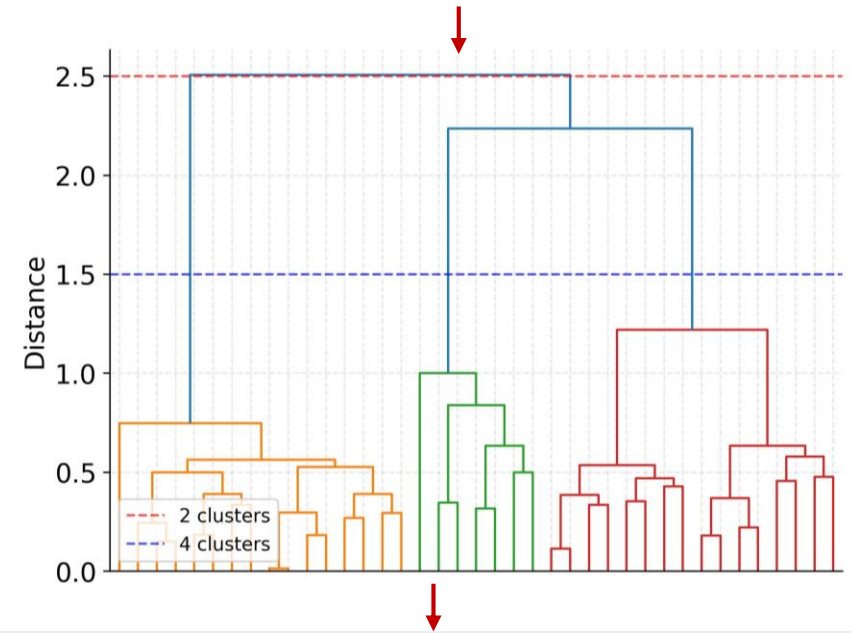
- Shows how observations/clusters are progressively merged.
- Leaves represent individual observations
- Branches show cluster mergers
- Height represents the distance/dissimilarity at which clusters are merged

Interpreting cluster relationships

- Clusters joined at a low height are more similar.
- Clusters joined at a high height are more dissimilar.
- The dendrogram reveals relationships at different levels of similarity.

Choosing a Cut Height

A **horizontal cut** across the dendrogram determines the resulting clusters



Effect of cut height

Higher cut → fewer, larger clusters.

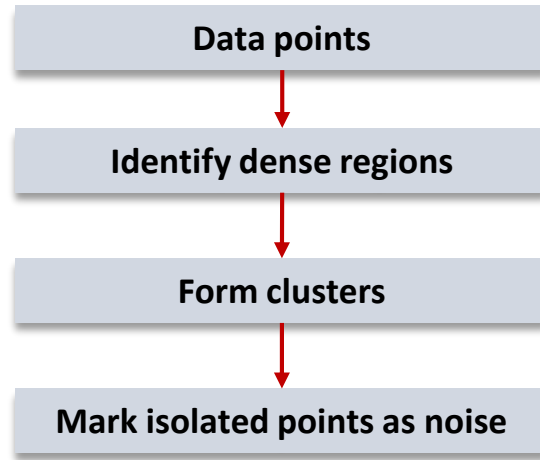
Lower cut → more, smaller clusters.

Different cut heights produce different numbers of clusters.

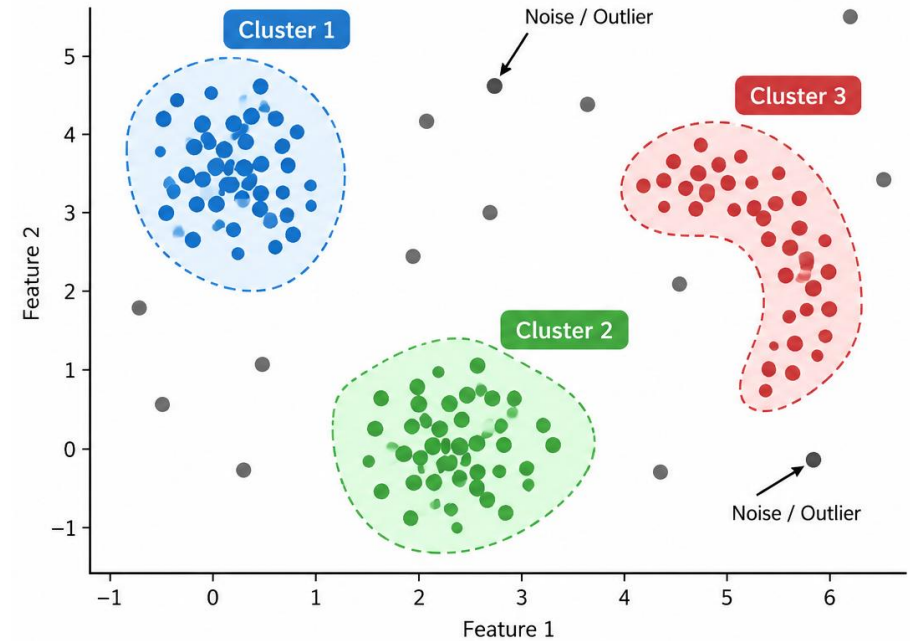
DBSCAN

- ▶ DBSCAN groups observations based on dense regions of data rather than distances from cluster centroids.

How it works — conceptually



- Dense region → observations are close together.
- Cluster → connected region of high data density.
- Noise / outlier → isolated observation that does not belong to a dense region.
- Can discover clusters of different shapes.



Important

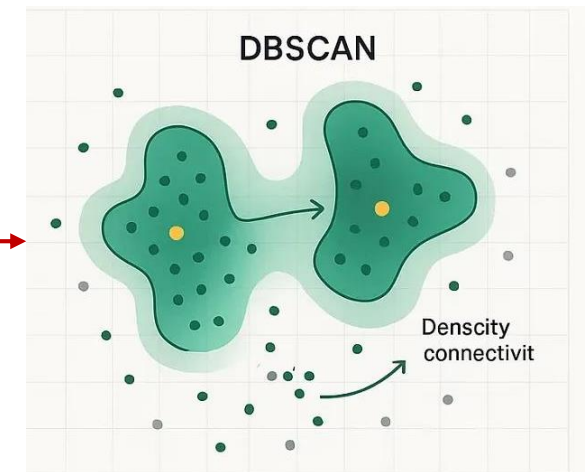
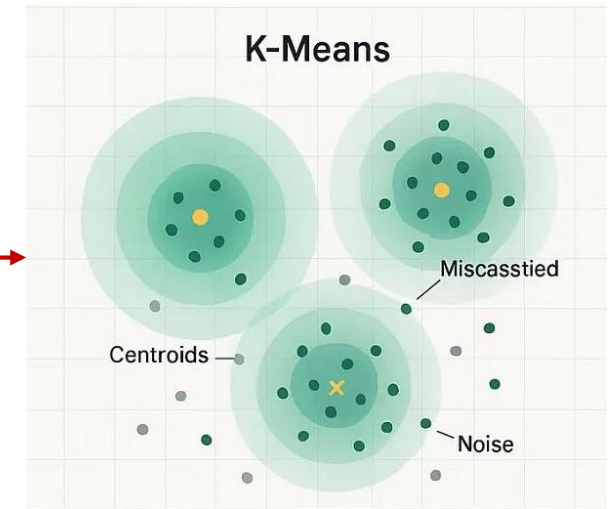
Unlike K-Means, DBSCAN does not organize clusters around centroids

DBSCAN vs Centroid-Based Clustering

	K-Means	DBSCAN
Main idea	Centroid-based	Density-based
Cluster structure	Centroid-oriented	Dense regions
Cluster shape	Roughly circular	Can be irregular
Noise / outliers	Can affect clusters	Can be identified as noise
Number of clusters	K must be chosen	No K specified beforehand

K-Means - Which centroid is this point closest to?

DBSCAN - Is this point part of a dense region?

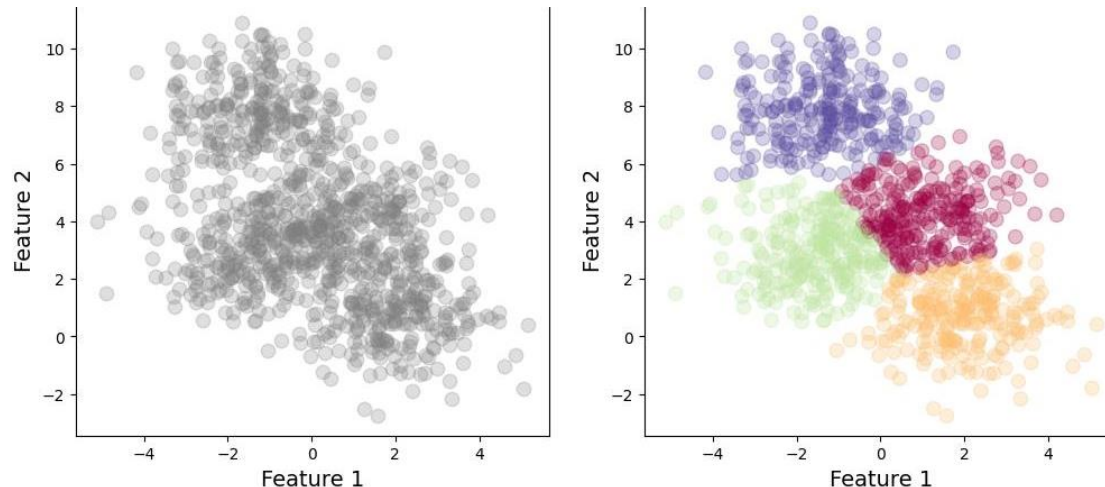


Cluster Validation

- ▶ A clustering algorithm will produce clusters - but are those clusters meaningful?

Why validation is needed

- Clusters may represent real structure in the data.
- They may instead be a result of noise or arbitrary choices.
- Different clustering methods or settings can produce different solutions.
- Therefore, clustering results should not automatically be treated as “true” groups.



Comparing Clustering Solutions

Try different

- K values in K-Means
- Linkage methods in hierarchical clustering
- Clustering algorithms

Alternative Clustering Solutions

Compare Results

Meaningful /
Interpretable?

Consider solution

Try another solution

Decision
concept

Cluster Interpretation

- ▶ A cluster label does not automatically tell us what the group means.

Interpret the characteristics

- Examine the features/variables within each cluster
- Identify characteristics that distinguish one cluster from another
- Describe the cluster based on its observed characteristics

Be cautious

Do not assume that every cluster represents a real-world group
Treat clustering results as a starting point for further investigation
Interpret clusters using the domain/business/research context

From Clusters to Meaningful Segments

What makes each cluster different



Is the difference meaningful



Does it relate to the business/research question



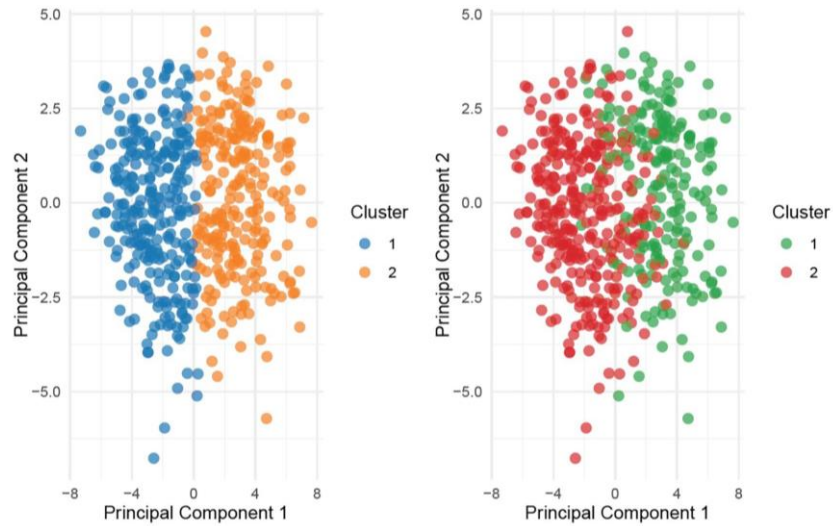
Can the cluster be given a meaningful interpretation

Key takeaway

Clustering discovers groups
Interpretation explains what those groups may represent

Comparing Clustering Results

K-Means	Hierarchical
Requires K to be specified	Produces a dendrogram
Centroid-based	Hierarchical merging
Produces a fixed partition	Can obtain different clusterings by cutting the dendrogram
Sensitive to initialization and choice of K	Sensitive to distance, linkage and cut height



Different methods can produce different cluster assignments from the same data

When comparing clustering solutions

1. Stability

Do similar clusters appear when the analysis is repeated or data is slightly changed

2. Interpretability

Can the resulting clusters be meaningfully explained in the context of the problem

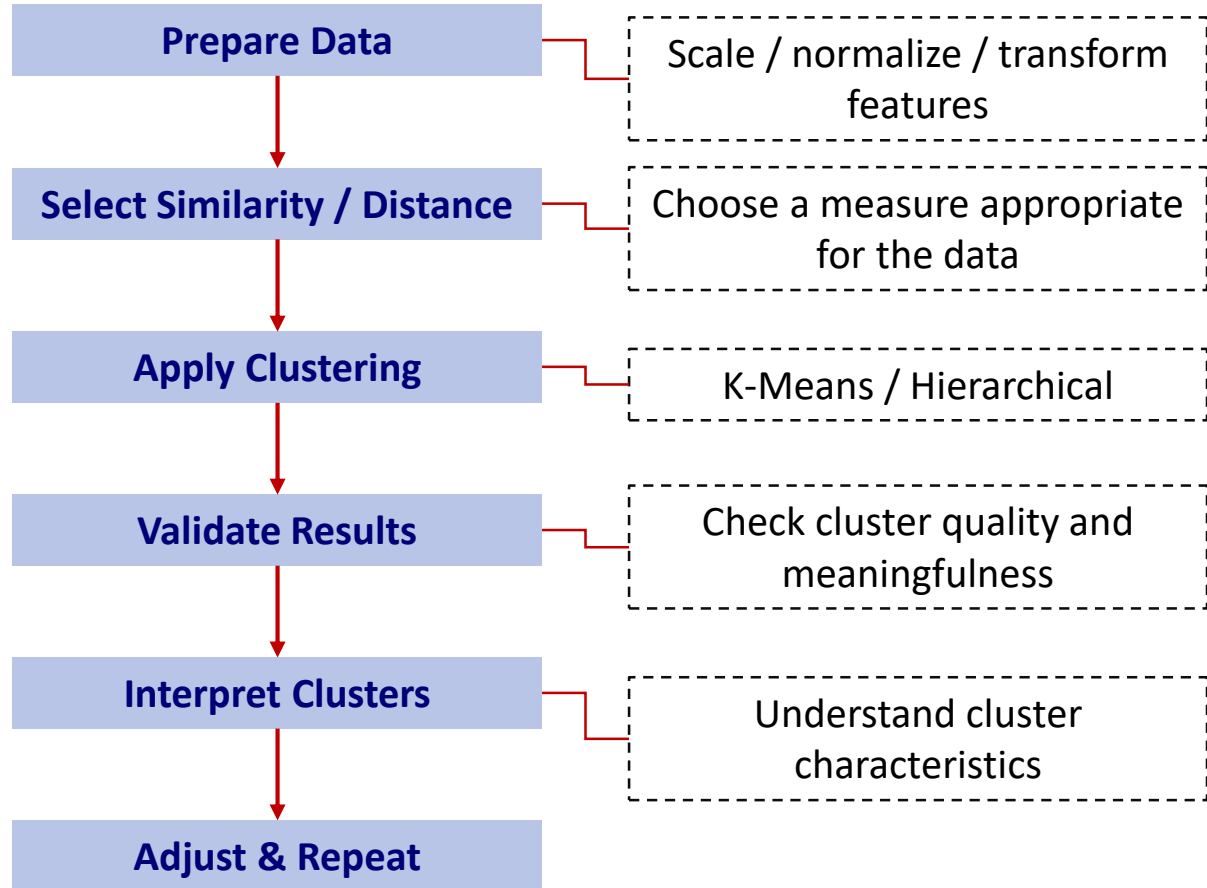
3. Usefulness

Does the solution reveal useful structure in the data

Key takeaway

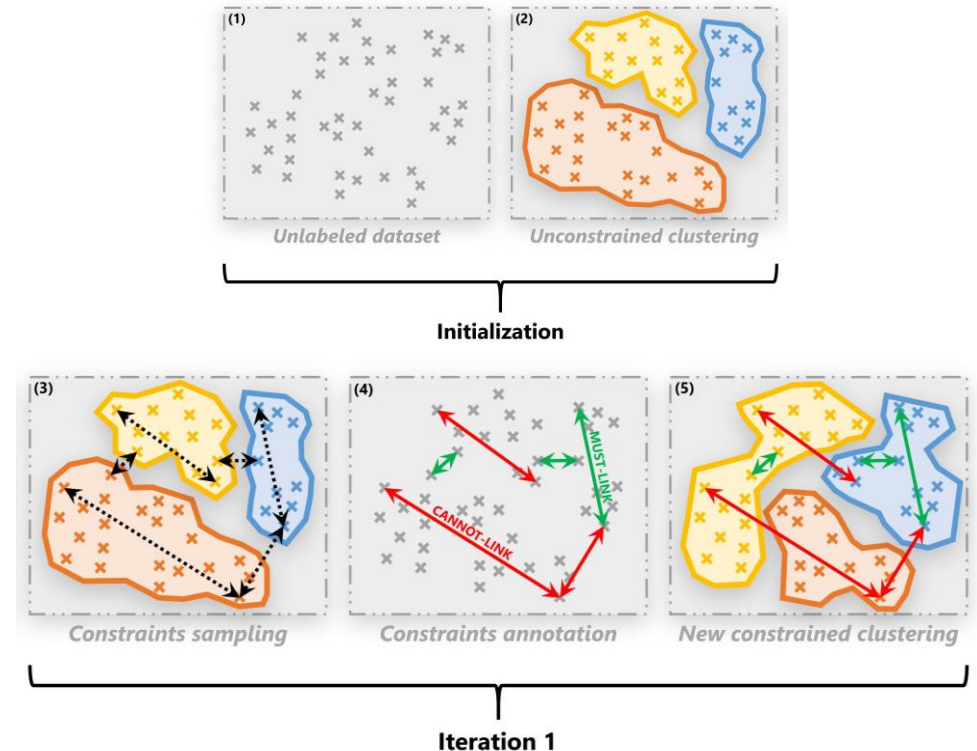
Do not choose a clustering result only because an algorithm produced it. Compare alternative solutions for stability, interpretability, and usefulness.

Practical Clustering Workflow



What to check

- Are similar observations grouped together?
- Is the chosen distance/similarity measure appropriate?
- Are the clusters meaningful and interpretable?
- Does the result meet the purpose of the analysis?



Summary

In this lecture, we learned:

- ▶ Unsupervised learning and its role in discovering patterns without a target variable.
- ▶ Clustering fundamentals, including similarity, distance measures, and feature scaling.
- ▶ K-Means clustering, including centroids, initialization, cluster assignment, and choosing K.
- ▶ Hierarchical clustering, including agglomerative clustering, dendrograms, linkage methods, and cut heights.
- ▶ DBSCAN, including density-based clustering, dense regions, arbitrary-shaped clusters, and noise/outlier identification.
- ▶ Cluster validation and the importance of checking whether discovered groups represent meaningful structure.
- ▶ Cluster interpretation by examining characteristics and relating them to the problem context.
- ▶ Comparison of clustering results using K-Means, hierarchical clustering, and DBSCAN.
- ▶ Practical clustering workflow from data preparation and distance selection to clustering, validation, interpretation, and adjustment.



Thank You